Lec-11

10/10/26 10:10 AM

0 1 2 3 4 5 6 7 8

```
int[] arr = { 3, 5, 6, 2, 4, 16, 7, 8, 9 };
int item = 16;
```

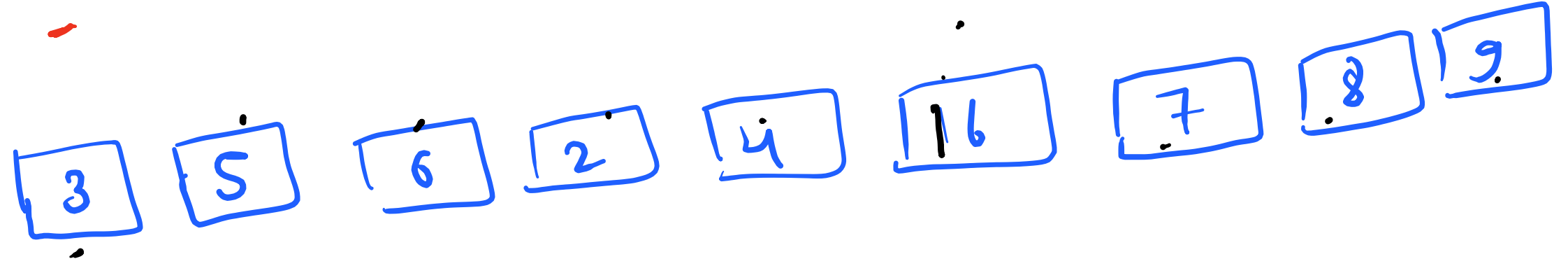
-1

SH

Pen →

```
for(i=0 i<N i++){
    if(arr[i]==item){
        retn i
    }
}
ret-1
```

```
int[] arr = { 3, 5, 6, 2, 4, 16, 7, 8, 9 };
```



max= ~~3~~ ~~5~~ ~~6~~ 16

~~3~~ ~~8~~ ~~5~~ 16

5>3

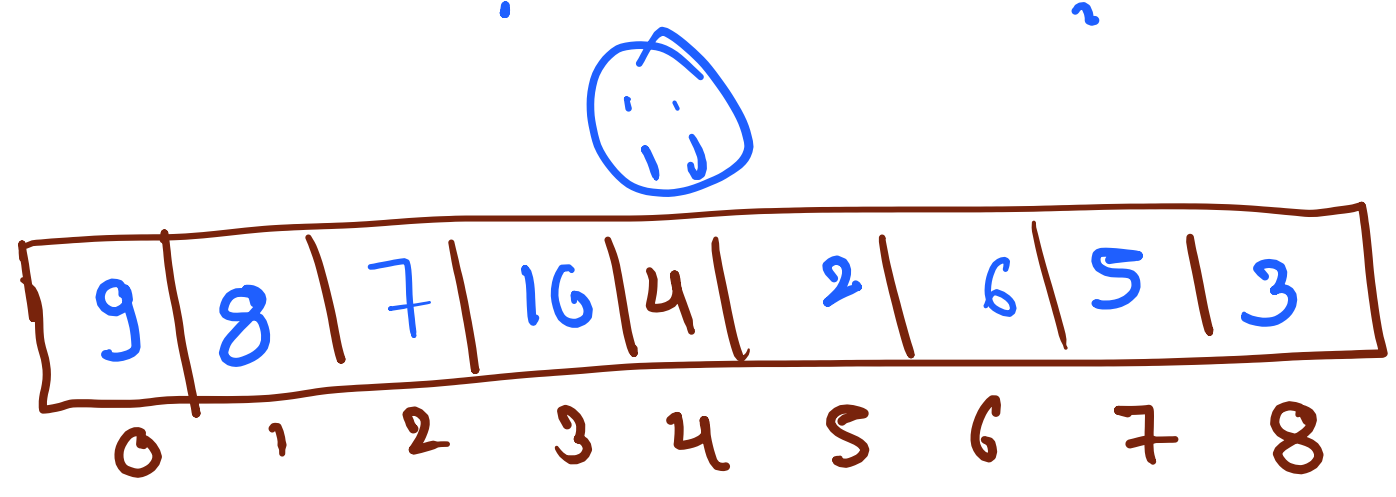
6>5

16>6

```
int max=arr[0]
for(i=1 i<N i++){
    if (arr[i]>max){
        max=arr[i]
    }
}
```

```
max=-∞
for(i=0 i<n i++){
    if(arr[i]>max){
        max=arr[i]
    }
}
```

```
int[] arr = { 3, 5, 6, 2, 4, 16, 7, 8, 9 };
```



3 5 6 2 4 16 7 8 9

9 8 7 6 1 4 2 6 5 3

i=0

j=arr.len-1

```
while(i<j){
    swap(i,j)
    i++
    j--
}
```

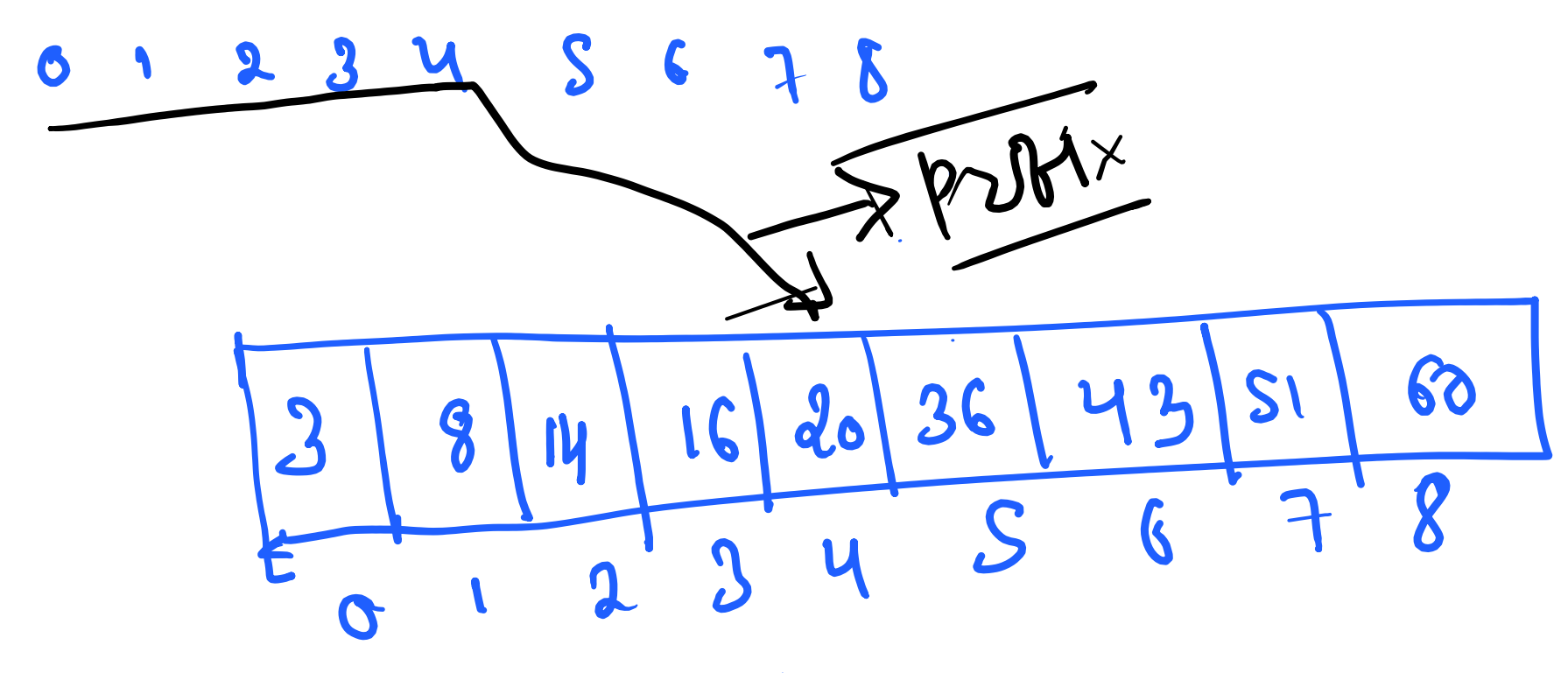
(0-8)

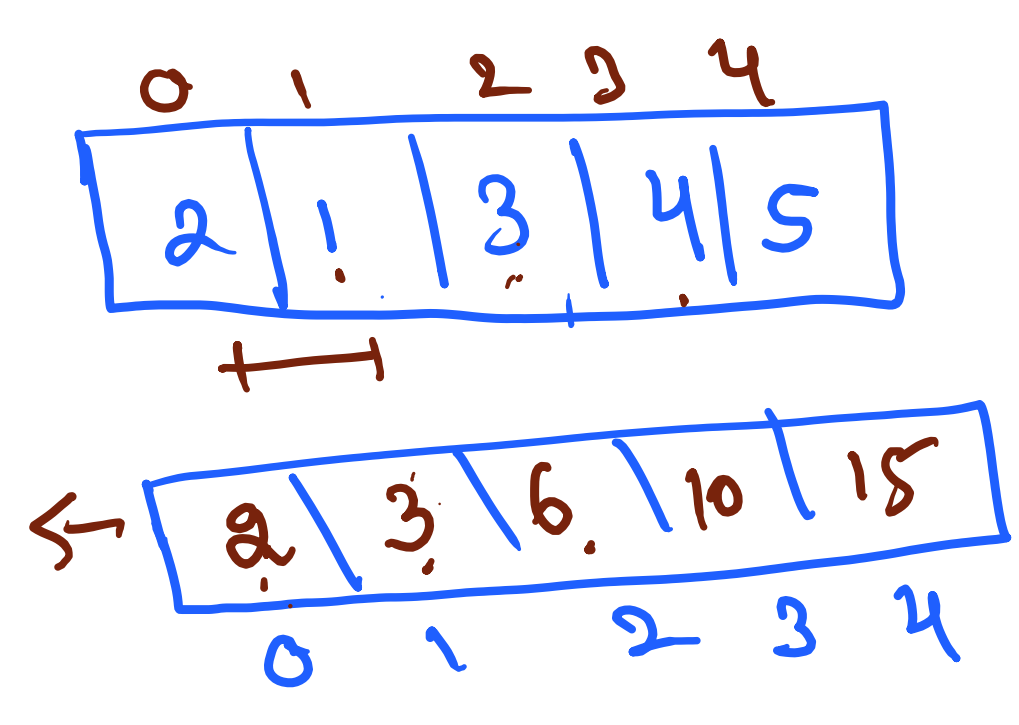
(1-7)

2-6

3-5

```
{ 3, 5, 6, 2, 4, 16, 7, 8, 9 };
```





left[0]=arr[0]

i=1 left[1]=left[0]+arr[i]

i=2 left[2]=left[1]+arr[2]

i=3 left[3]=left[2]+arr[3]

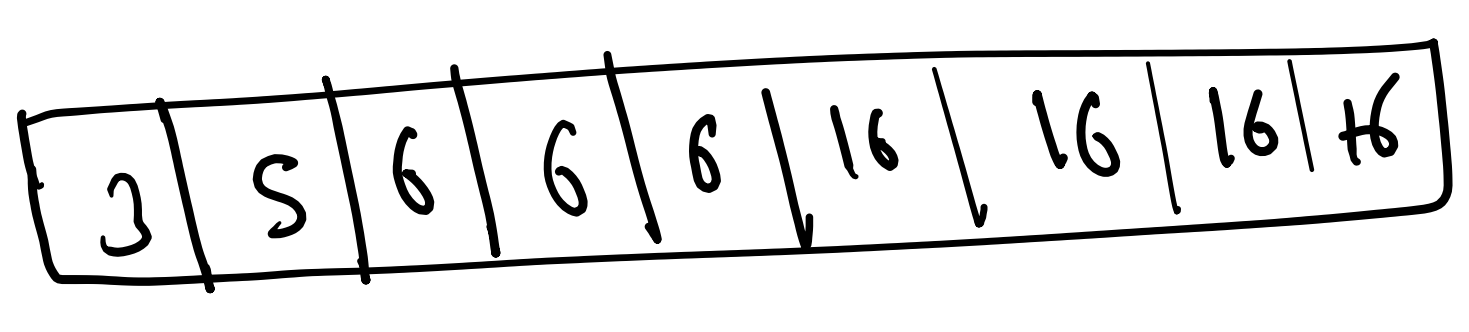
i=4 Left[4]=left[3]+arr[4]

left[i]=left[i-1]+arr[i]

```
for(i=1 i<n i++){
    left[i] =left[i-1] +arr[i]
}
```

```
int[] arr = { 3, 5, 6, 2, 4, 16, 7, 8, 9 };
```

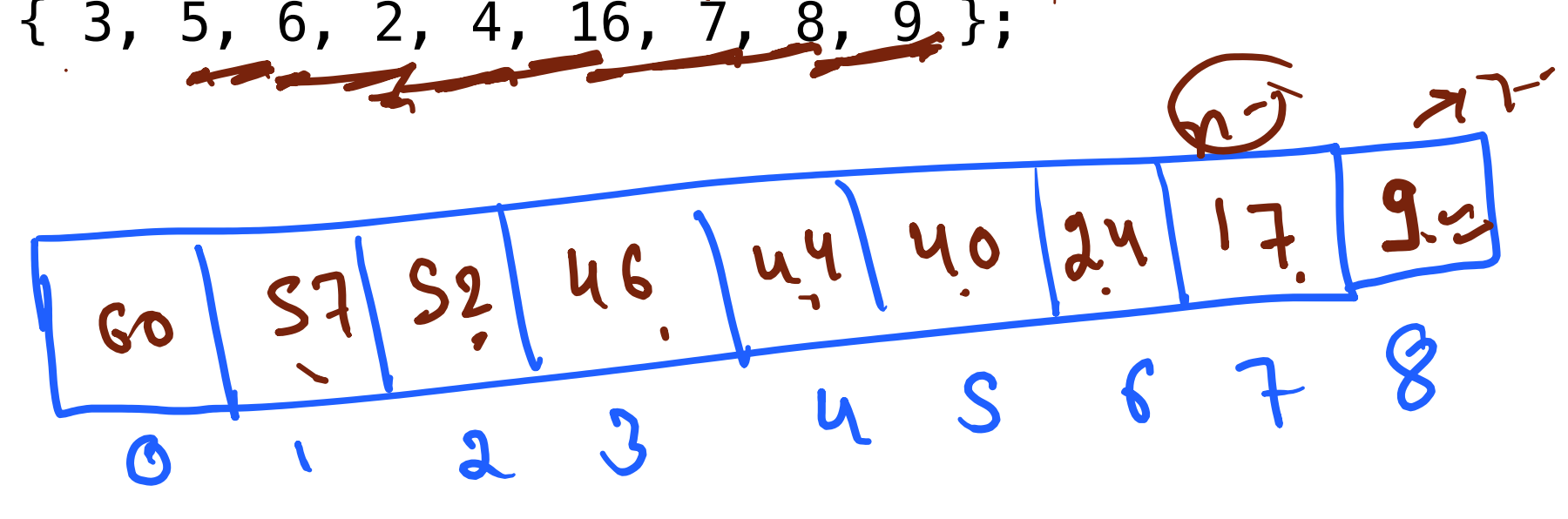
Prefix



Suffix

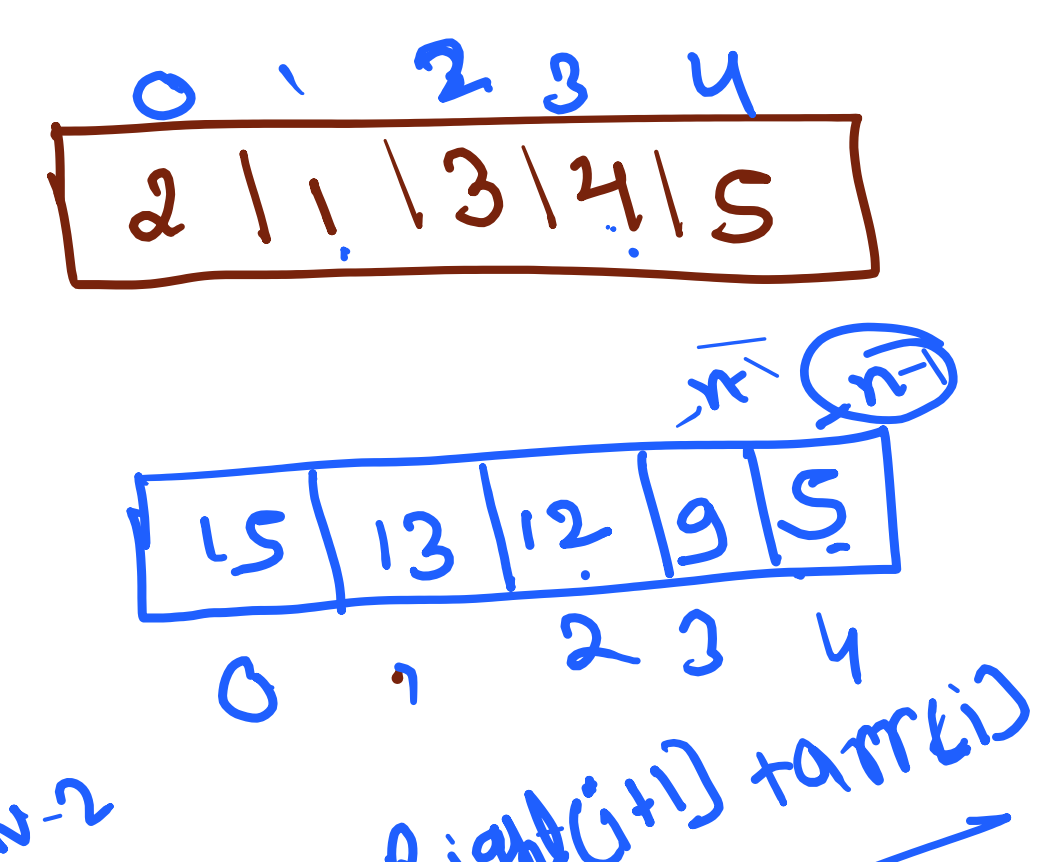
```
int[] arr = { 3, 5, 6, 2, 4, 16, 7, 8, 9 };
```

N=arr.len



R CN

Right [N-1] =arr[N-1]



N=arr.len

① Right[N-1]=arr[N-1]

i=3 Right[3]= Right[4]+arr[3]

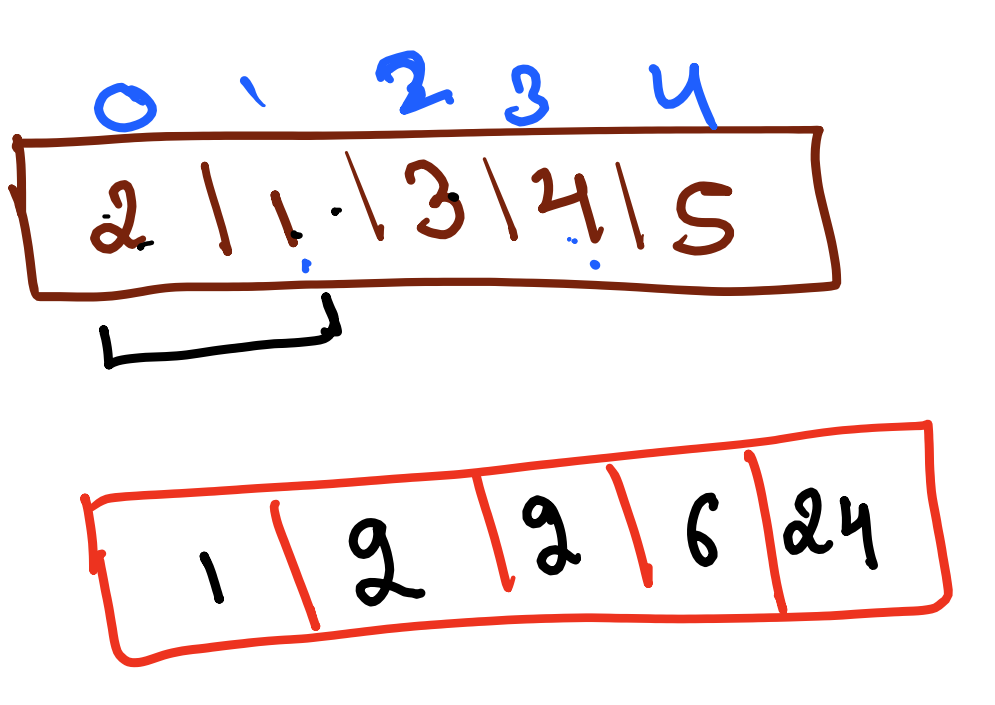
i=2 Right[2]=Right[3]+arr[2]

i=1 Right[1]= Right[2]+arr[1]

i=0 Right[0]= Right[1]+arr[0]

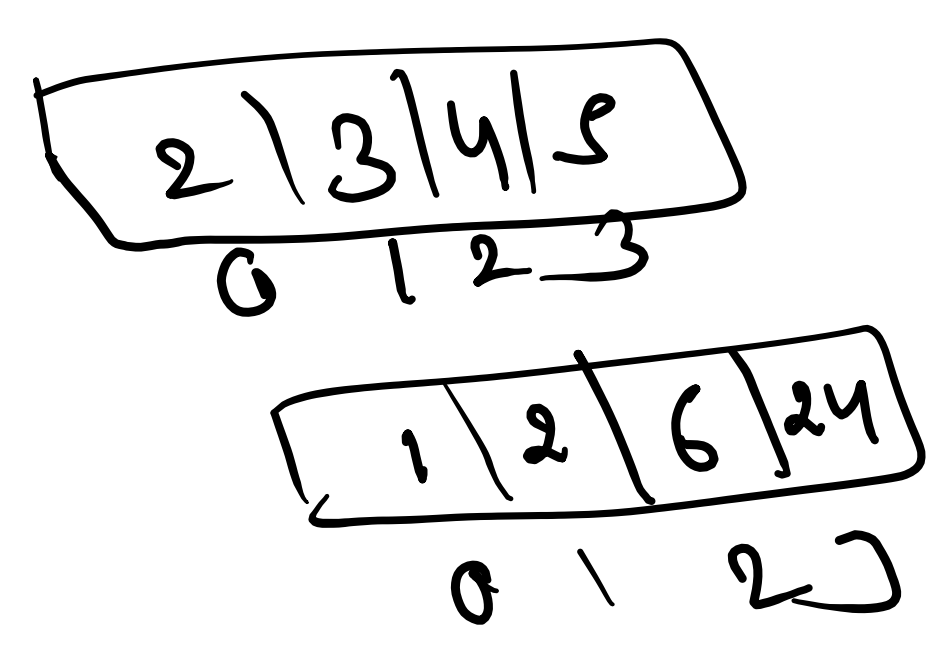
i=N-2

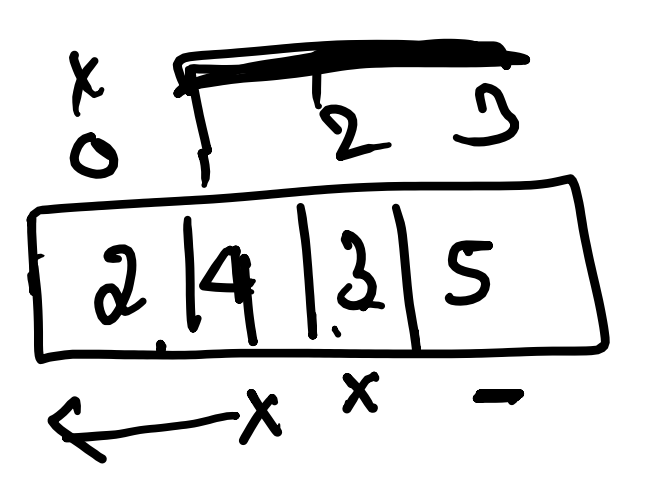
Right[i]= Right[i+1] +arr[i]



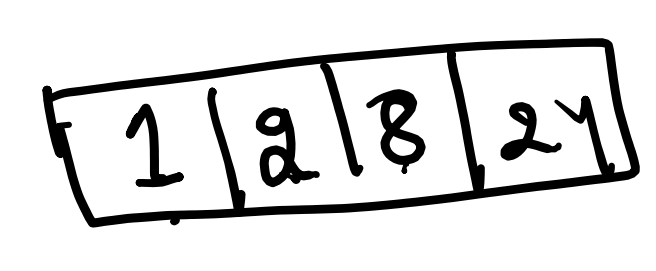
Prefix

↳ Left





Left [0] = 1



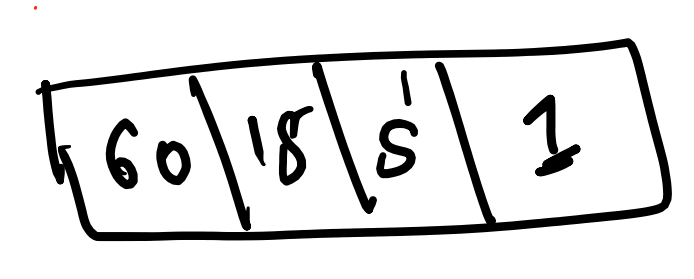
Left [1] =Left[0] xarr[0]

Left [2]= Left[1] xarr[1]

Left [3]= Left[2] Xarr[2]

Left[i]=Left[i-1]*arr[i]

suffix



R[i]=R[i+1]xarr[i+1]

Right [3]= 1

i=2 Right [2]= Right[3] xarr[3]

i=1 Right[1] =Right[2] x arr[2]

i=0 Right [0]=Right[1]xarr[1]

[2,5,4,1,7,6] 2nd largest element

```
int[] arr = { 2, 5, 4, 1, 7, 6 };
```

first = arr[0] → ~~2~~ ~~5~~ 7

Second = arr[0] → ~~2~~ ~~2~~ ~~4~~ ~~5~~ 6

```
for (i=1 i<N i++){
    if (arr[i]>first){
        sec = first
        first = arr[i]
    }
    else if (arr[i] > sec){
        sec = arr[i]
    }
}
```